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Question Bank for MSC IT
Semester – VII

Subject code:	50020701
Subject Name:	Analysis and Design of Algorithms
Prepared By:	Prof. Ritisha Parmar

Unit 1: Basic Concepts of Algorithms and its Analysis	
Unit	Questions
1	Define an algorithm. List various criteria used for analyzing an algorithm
2	Arrange the following rate of growth in increasing order 2n , n log n, n2 ,1,n, log n, n! , n3
3	Define Recursion. Give the limitations of recursion. How to compute the time complexity of recursive algorithms?
4	Define order notation. What is the use of it? What are the various order notations are available? Explain any one of them with example
5	What is an algorithm? How it differ from flowchart?
6	Define Algorithm. Discuss key characteristics of algorithm
7	Find out the Θ-notation for the function: f(n)=27n2+16n
8	Why do we use asymptotic notations in the study of algorithms? Briefly describe the commonly used asymptotic notations.
9	What is the smallest value of n such that an algorithm whose running time is 100n2 runs faster than an algorithm whose running time is 2n on the same machine?
10	1. What is an algorithm? 2 What is worst case time complexity ? 3 Define space complexity. 4 Define Big Omega Asymptotic Notation
11	Write down the Best case, Worst Case and Average case Complexity for merge sort
12	Write down the Best case, Worst Case and Average case Complexity for selection sort.
13	Write down the Best case, Worst Case and Average case Complexity for Heap sort
14	Worst complexity of an algorithm is represented by omega notation. (True/false) ANS: “big O is used for worst case, Theta for average case
15	Which of the algorithm having following time complexity will take the highest amount of time to complete? O(n2) , O(log n), O(xn)
Unit 2: Algorithms using Divide and Conquer	
Unit	Questions

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1	Write a note on Strassens matrix multiplication algorithm. Solve the following using strassens matrix multiplication algorithm and find A X B. A= 3 4 5 6 B= 2 3 1 4
2	Explain difference between divide and conquer method and dynamic programming. Explain Stressen's Multiplication and compare the time Complexity of that algorithm with regular matrix multiplication.
4	What do you understand by Divide and conquer strategy? Explain the precise for of D & C algorithms. Also explain the Binary Search algorithm.
5	Write an algorithm to find the power of a given number using divide and conquer strategy
6	Explain how multiplication of large integers can be done efficiently by using divide and conquer method.
7	Multiply 981 by 1234 by divide and conquer method.
8	Explain Divide and Conquer strategy. What happens in D & C? Write the algorithm for multiplying two large numbers using D & C and give its time complexity. Give the tracing for 2431 X 2212
9	What is recurrence? Solve recurrence equation T (n) =T (n-1) + n using forward substitution and backward substitution method
10	What do you mean by Divide & Conquer approach? List advantages and disadvantages of it.
11	Solve the following recurrence relation using iteration method. T(n) = 8T(n/2) + n2 . Here T(1) = 1.
12	Solve the following recurrence relation using substitution method. T(n) = 2T(n/2) + n. Here T(1) = 1
13	Solve the recurrence:$T(n)=7T(n/2)+n^3$
14	Solve following recurrence using recursion tree method: T(n) = 3T(n/3) + n^3.
15	Solve following recurrence using master method T(n) = 9T(n/3) +n
16	Solve following recurrence using master method T(n) = T(2n/3) + 1
17	What is recurrence? Explain recursion-tree method with suitable example.
18	Write the Master theorem. Solve following recurrence using it. (i) T(n)=9T(n/3) + n (ii) T(n)=3T(n/4) + nlgn
19	What is hiring problem write it with randomized algorithm?
20	Explain quick sort algorithm

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21	Explain 2-way merge sort algorithm
22	Explain maximum subarray algorithm with psuedocode.
Unit 3: Algorithm solving using Dynamic Programming	
Unit	Questions
1	Compare and contrast – Divide and Conquer, Greedy and Dynamic problem solving algorithms.
2	Solve the following rod cutting problem. Rod size is 6 Profit for rod of sizes {1,2,3,4,5,6} prices= {5,2,4,1,3,6}
3	Explain the rod cutting problem. Discuss its time complexity
4	Suppose the user wants to purchase 5 m length of rod and for that he went to the rod seller. Seller having 4 rods with the size 1, 2, 3 and 4 and each having profit 2, 5, 9, 8. Now, provide the solution for selling the product in such way that the user gets the rod of length 5 and the seller gets the maximum profit.
5	Using algorithm find an optimal parenthesization of a matrix chain product whose sequence of dimension is (5,10,3,12,5,50,6)(use dynamic programming).
6	Using algorithm determine an Longest Common Sequence of (A,B,C,D,B,A,C,D,F)and (C,B,A,F)(use dynamic programming).
7	Explain how to find out Longest Common Subsequence of two strings using Dynamic Programming method. Find any one Longest Common Subsequence of given two strings using Dynamic Programming. X=abbacdcba Y=bcdbbcaac
8	Find out LCS of A={K,A,N,D,L,A,P} AND B={A,N,D,L}
9	Determine LCS of {1,0,0,1,0,1,0,1} and {0,1,0,1,1,0,1,1,0}
10	Discuss the longest common sub sequence with following example and provide the longest common sub sequence table. S1=ABCB S2=BDCAB
11	Explain Matrix Chain Multiplication with example
12	Write a note on Matrix chain multiplication algorithm. Solve the following using dynamic programming approach. Find out the order in which the following matrices should be multiplied so to reduce the total number of multiplications involved. A[10,15] X B[15,12] X C[12,11] X D[11,20]
13	Find optimal sequence of multiplication using dynamic programming of following matrices: A1[10x100], A2[100x5], A3[5x50] and A4[50x1]. List optimal number of multiplication and parenthesization of matrices.
14	For the following chain of matrices find the order of parenthesization for the optimal chain multiplication (15,5,10,20,25)
15	For the following chain of matrices find the order of parenthesization for the optimal chain multiplication(13,5,89,3,34)
16	Using dynamic programming find out the optimal sequence for the matrix chain multiplication of A4x10, B10x3, C3x12, D12x20 and E20x7 matrices

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17	Write equation for Chained matrix multiplication using Dynamic programming. Find out optimal sequence for multiplication: A1 [5 × 4], A2 [4 × 6], A3 [6 × 2], and A4 [2 × 7]. Also give the optimal parenthesization of matrices
18	Which are the three basic steps of the development of the dynamic programming algorithm? Mention any two examples of dynamic programming that we are using in real life
19	What do you understand by binary search trees. Explain with example optimal binary search trees.

Unit 4: Greedy Algorithms

Unit	Questions
1	Write a short note on Activity selection problem using greedy approach.
2	What do you understand by Huffman code? For what it is used? Explain with appropriate example.
3	Write a note on how Huffman code is generated using greedy approach.
4	Explain the elements of greedy strategy.
5	Generate the Huffman tree for the following data Characters - a b c d e f g h i Frequency - 9 12 10 1 20 6 18 5 25
6	Explain job sequencing with deadline algorithm
7	Solve the following job sequencing problem P={23,20,15,16} and D={1,3,2,1}
8	Write a note on spanning trees
9	Explain prim's algorithm with example
10	Explain kruskal's algorithm with example
11	Explain dijikstra's algorithm
12	Explain BellmanFord Algorithm

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Unit 5: Algorithms using Branch and Bound and Backtracking method	
Unit	Questions
1	Explain what do you understand by backtracking using n-queens problem example
2	Write a note on DFS algorithm
3	Write a note on BFS algorithm
4	What is 8-puzzle problem?
5	Explain knapsack problem using branch and bound strategy
6	Differentiate between back tracking and branch and bound strategy
7	Differentiate between DFS and BFS
8	DFS and BFS problems on graphs for solving
Unit	**Questions**
1	Define the following questions: i. Define Time complexity. ii.Define Feasible Solution. iii.Define P-type Problem. iv.Define Algorithm. v. Define Minimum Spanning Tree. vi. Write down the Best case, Worst Case and Average case Complexity for Heap sort. vii. Define Optimal Solution.
2	Do as Directed 1. The Complexity of merge sort is [O(n),O(log n),O(n log n),O(n2)] 2. What are the two main measures for the efficiency of an algorithm? 3. Define Algorithm. 4. Define Optimal Solution and Feasible Solution. 5. What is the complexity to add and delete element from bit vector. 6. Define Directed Acyclic graph. 7. Define Minimum Spanning Tree.

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3	Answer the following: i) Define an algorithm. ii) Write the complexity of binary search algorithm. iii) List out the algorithm for solving minimum spanning tree. iv) Define multi stage graph. v) What is state space in backtracking? vi) Distinguish NP and P problem. vii) Which data structure is use for kruskal's algorithm.

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